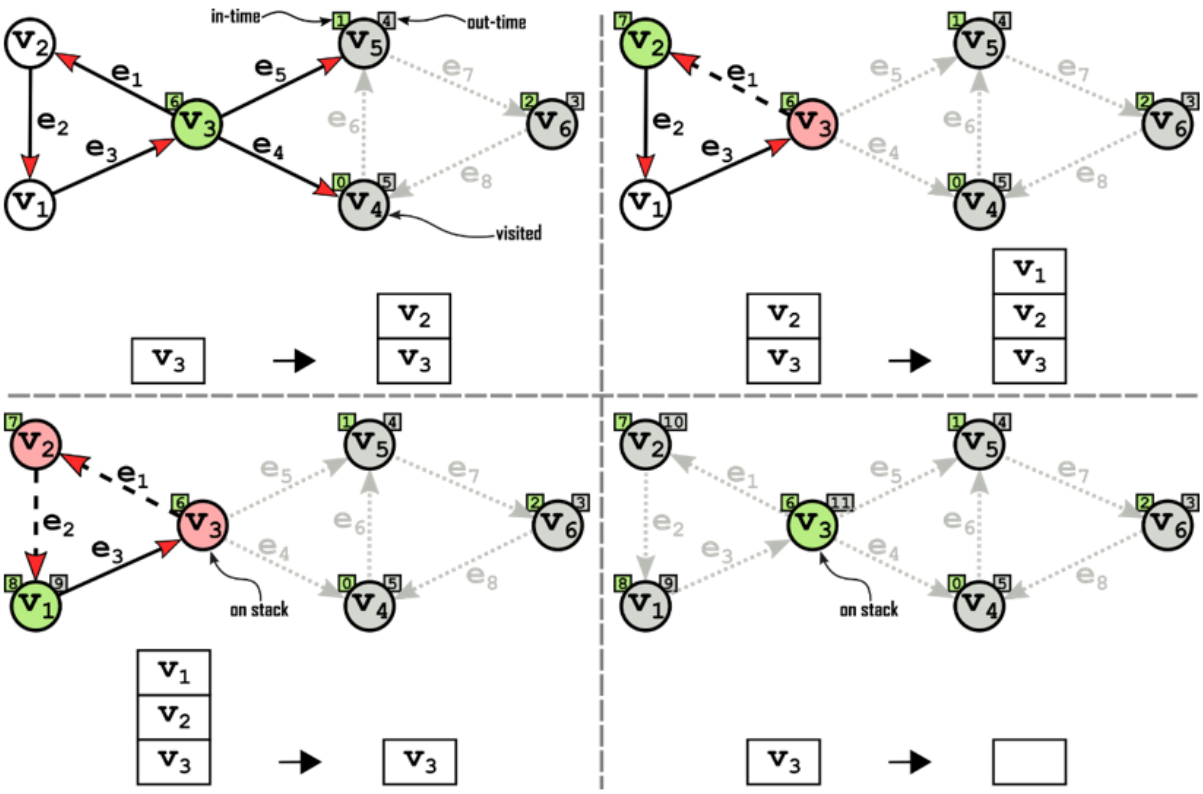


Figure 14.13 DFS in action: following up on the example in figure 14.10, we can resume the traversal and get to all the vertices by randomly selecting one of the vertices left, and re-starting DFS from it.

A final note on performance: as for BFS, this algorithm also has a linear running time $O(|V| + |E|)$, and requires $O(|V|)$ recursive calls (or, equivalently, $O(|V|)$ extra space for the stack).

14.3.5 It's Queue vs Stack Again

When we look at these traversal algorithms, the first thing that should be clear is that their purpose, the context in which they are applied, is fundamentally different: BFS is used when the source vertex S is known, and we want to find